

ROGERS CITY, MI, USA

WMO: 727417

Lat: 45.407N

Lon: 83.813W

Elev: 670

StdP: 14.34

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 54825

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.1	3.4	-6.9	3.9	1.3	-3.1	4.8	5.1	22.7	21.4	20.4	22.0	7.9	260	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	86.0	69.3	82.2	67.8	79.3	66.8	72.5	81.2	70.7	78.4	69.1	76.0	9.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.9	112.6	77.4	68.1	106.0	75.3	66.3	99.4	73.5	36.6	81.1	34.9	79.1	33.6	76.1	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.1	18.1	16.3	-6.1	92.1	5.4	2.5	-10.0	93.9	-13.2	95.4	-16.3	96.8	-20.2	98.6		
			-6.7	76.1	5.3	1.5	-10.5	77.2	-13.6	78.1	-16.6	78.9	-20.5	80.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.8	21.4	21.6	29.5	40.5	52.6	61.9	68.1	66.7	60.0	48.9	37.1	27.5
	DBStd	18.60	9.89	9.78	10.45	8.51	8.71	7.05	5.77	5.51	7.84	8.54	8.57	8.22
	HDD50	3929	886	796	641	307	68	2	0	0	7	127	397	698
	HDD65	7717	1351	1216	1100	736	399	141	34	45	190	504	838	1163
	CDD50	2025	0	0	7	21	149	360	561	516	308	94	9	0
	CDD65	340	0	0	1	1	15	50	130	96	41	6	0	0
	CDH74	2285	0	0	13	13	158	369	876	547	268	41	0	0
	CDH80	575	0	0	3	2	48	104	217	123	72	6	0	0
	WSAvg	7.5	8.2	8.7	8.2	8.3	6.9	6.1	6.1	6.2	6.7	7.8	8.1	8.3
	PrecAvg	30.0	2.1	1.5	1.6	2.3	2.9	3.1	3.3	3.4	3.1	2.6	2.4	2.0
(15) Precipitation	PrecMax	34.4	3.9	3.1	4.9	3.6	6.7	6.1	6.2	7.0	6.3	6.1	4.3	5.0
	PrecMin	23.6	0.9	0.5	0.1	0.6	0.7	0.4	1.3	1.3	1.3	0.5	0.9	0.5
	PrecStd	3.0	0.9	0.8	1.0	0.9	1.3	1.4	1.4	1.6	1.3	1.3	0.9	1.1
	PrecStd	3.0	0.9	0.8	1.0	0.9	1.3	1.4	1.4	1.6	1.3	1.3	0.9	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.0	49.7	70.7	73.5	86.1	89.2	90.1	88.9	87.9	79.5	66.7	52.6
		MCWB	42.5	44.8	57.4	56.7	65.5	70.3	71.9	70.8	69.9	63.7	54.9	49.0
	2%	DB	41.1	41.2	54.9	65.8	78.2	82.3	85.4	82.9	81.3	72.1	58.0	44.9
		MCWB	37.9	37.3	47.4	55.3	62.6	68.1	70.3	68.4	67.6	63.1	50.5	41.6
	5%	DB	37.1	37.0	48.1	59.1	72.5	78.6	81.9	79.5	76.5	66.5	53.7	40.9
		MCWB	34.9	33.9	41.5	49.4	60.1	65.8	68.3	67.7	66.0	59.7	48.1	38.2
	10%	DB	34.6	34.2	43.1	53.9	67.9	73.4	79.1	76.8	72.6	62.5	49.7	37.9
		MCWB	33.1	31.8	37.7	44.9	58.0	63.2	66.9	66.5	64.5	56.9	45.4	35.8
	0.4%	WB	44.2	44.3	59.1	59.6	69.4	73.3	75.2	73.4	73.4	68.7	57.3	49.8
		MCDB	45.4	48.5	70.5	68.5	80.4	84.1	84.9	81.0	82.9	75.7	64.2	50.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.4	37.9	49.0	54.8	66.0	70.2	72.7	71.5	70.4	64.5	52.0	42.4
		MCDB	40.6	41.0	54.6	65.8	73.6	79.1	81.3	79.6	77.1	68.6	57.2	44.3
	5%	WB	35.3	34.3	42.0	50.0	62.4	67.7	70.7	69.9	68.1	60.5	48.5	38.6
		MCDB	36.8	36.4	47.3	57.8	70.8	75.1	78.4	76.2	74.4	65.9	52.9	40.4
	10%	WB	33.1	32.1	37.9	45.9	58.8	65.5	68.9	68.3	65.8	57.0	45.7	36.1
		MCDB	34.6	33.9	42.3	52.5	66.8	73.0	76.3	74.5	72.0	61.9	48.7	38.0
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	13.0	14.6	16.2	19.9	18.7	17.9	16.8	17.2	14.2	11.7	9.5
		MCDBR	12.4	14.7	22.1	26.3	26.9	23.1	21.6	20.7	21.9	22.0	18.9	13.5
		MCWBR	10.6	11.6	14.7	16.3	14.7	12.5	10.9	10.3	11.3	14.1	13.3	11.6
	5% WB	MCDBR	11.9	14.1	20.8	25.1	24.5	20.3	18.6	17.3	18.7	19.9	17.5	12.5
		MCWBR	10.6	12.0	14.9	16.6	15.5	12.7	10.7	9.7	11.6	14.5	13.6	11.5
(40) Clear-Sky Solar Irradiance	taub		0.294	0.295	0.313	0.353	0.372	0.401	0.399	0.387	0.365	0.335	0.305	0.281
	taud		2.360	2.342	2.409	2.354	2.381	2.331	2.354	2.386	2.439	2.500	2.525	2.466
	Ebn at Noon		258	282	292	288	284	275	274	274	271	265	256	254
	Edh at Noon		24	29	32	37	37	39	38	35	31	25	21	20
	RadAvg		380	690	1164	1481	1774	1930	1948	1653	1276	754	404	273
(45) All-Sky Solar Radiation	RadStd		39	67	110	141	161	147	72	88	113	82	57	28

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page